

Aleksandar Angelov Yanev

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EDUCATION

Sofia University St. Kliment Ohridski, Computer Science

GPA: 5.97/6.00

Sofia, Bulgaria

October 2023 – Current

HONORS AND AWARDS

- INSAIT Summer Undergraduate Research Fellowship 2024 (SURF)
- Interuniversity mathematics competition 2023, 2024 – Silver medal, Bronze medal
- Open Mathematical Olympiad for University Students 2024 (OMOUS) – Silver medal (ind.), Silver medal (team)
- National Undergraduate Mathematics Olympiad 2024 – Bronze medal
- National Undergraduate Informatics Olympiad 2024, 2025 – 2nd place, 3rd place
- International Autumn Tournament in Informatics 2019, 2022 – Bronze Medal
- Romanian Masters of Informatics 2022 – Bronze Medal
- International Zhautykov Olympiad in Computer Science 2022 – Bronze Medal

RESEARCH & WORK EXPERIENCE

INSAIT - Institute for Computer Science, Artificial Intelligence and Technology

Sofia, Bulgaria

Residence researcher in Computer Vision

July 2024 – Current

- Researching and developing offline evaluation strategies for VLA models using real-world demonstration data
- Creating inference-time prediction ensembles to improve model performance

Smartsoft Healthcare

Varna, Bulgaria

Junior Machine Learning Developer

June 2023 – April 2024

- Fine-tuning and optimizing the neural network for analysis of the lumbar spine developed by the company
- Integrating and leveraging partially annotated data to boost the model's performance
- Collaborating with machine learning experts and biologists to analyze experiments and incorporate insights from complex patient studies for model improvement

PUBLICATIONS

- **SPEAR-1: Scaling Beyond Robot Demonstrations via 3D Understanding** [Paper](#)
N. Nikolov, G. Albanese, S. Dey, [A. Yanev](#), L. V. Gool, J. Zaech, D. Paudel
in *Computer Vision and Pattern Recognition (CVPR)*, 2026
 - Created a pipeline for enriching 2D image data with 3D information which is used to improve the performance of the model
 - Creating inference-time optimizations by applying temporal smoothing for the trajectories, normalizing predictions through multiple sampling, and implementing similarity metrics to improve predictions with multiple action chunks
- **InTraGen: Trajectory-controlled Video Generation for Object Interactions** [Paper](#) | [Code](#)
Z. Liu*, [A. Yanev](#)*, A. Mahmood, I. Nikolov, S. Motamed, S. Zheng, D. Paudel
in *arXiv preprint Nov. 2024*
 - Developed synthetic datasets to enhance text-to-video generation models' understanding of physical interactions
 - Exploring diverse control methods to improve interaction comprehension in video generation
 - Working on advancing novel architectures for text-to-video generation models by incorporating the newly developed control methods, with an emphasis on achieving meaningful object interactions
- **Utilizing Autoencoders for Analysis and Classification of Microscopic in Situ Hybridization Images** [Paper](#)
[A. Yanev](#), G. Momcheva, and S. Pavlov
in *Proceedings of the Bulgarian Academy of Sciences*, 2023

EXTRACURRICULARS

- Teaching Algorithms in Competitive Programming to 10th and 11th graders to prepare them for national and international tournaments at Olympians' Academy since 2023.
- "Yale Book Award" recipient, Fenced in épée at "Cherno More", "Benjamin Franklin's Legacy Program" awardee, Ex-president of the Library and Student Councils at High School of Mathematics